

Evaluating Diffusion Models for Cosmological Simulation

Data: Memorization, Generalization, and Scientific Fidelity

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The question

Generative emulators promise **cheap surrogates** for expensive cosmological simulations — but an emulator that **memorizes** its training set produces confident nonsense when used for inference. We train diffusion models on CAMELS HI maps and ask:

1. When does the model stop **copying** training fields and start **sampling** the underlying distribution?
2. Does conditional generation actually **respond to the requested cosmology** θ ?
3. Are generated fields **statistically faithful** (one-point PDF, power spectrum)?

Data & models

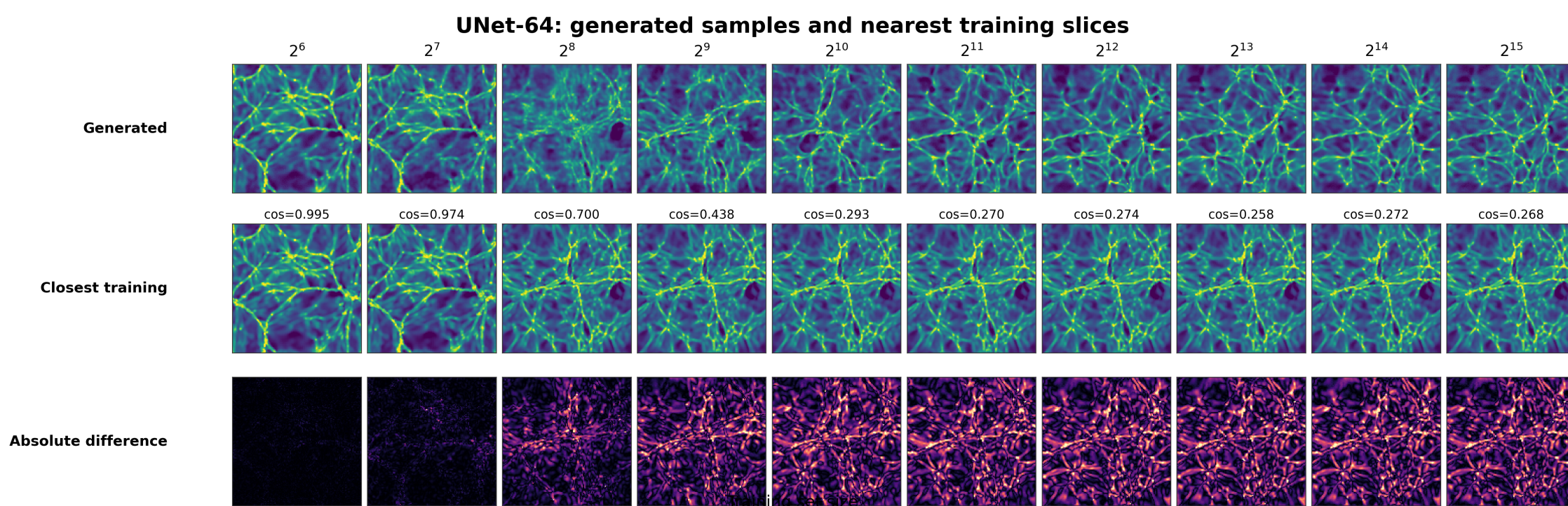
- **CAMELS** (IllustrisTNG) HI fields $\rightarrow 128 \times 128$ slices; 6 parameters $\Omega_m, \sigma_8, A_{SN1}, A_{AGN1}, A_{SN2}, A_{AGN2}$.
- **Diffusion**: UNet denoisers (widths 64 / 128 / 256); conditional variant injects θ at every level via cross-attention.
- **Sampling**: 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep**: $N = 2^6 \rightarrow 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

Detecting memorization

For each generated sample x_j , find its nearest training field in pixel, PCA, and SSCD embedding space:

$$s_j = \max_i \text{sim}(x_j, y_i), \quad \text{GL}(\tau) = 1 - \Pr[s_j > \tau]$$

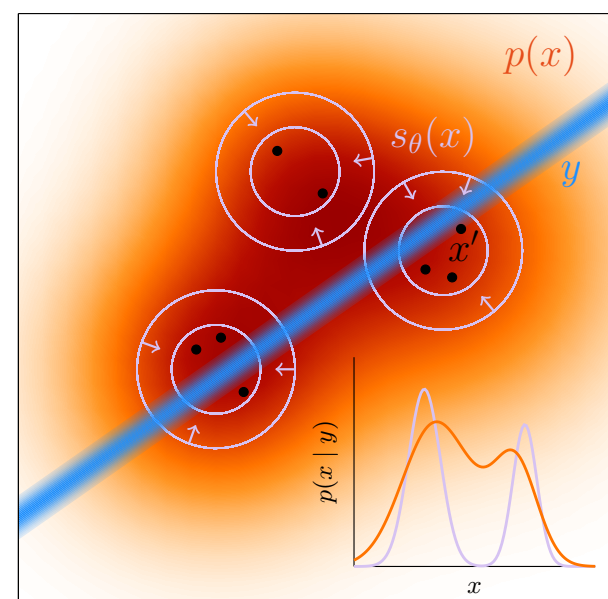
with τ calibrated from leave-one-out train–train similarities. Independent models should reproduce the same distribution, not the same training images.



Why this matters for inference

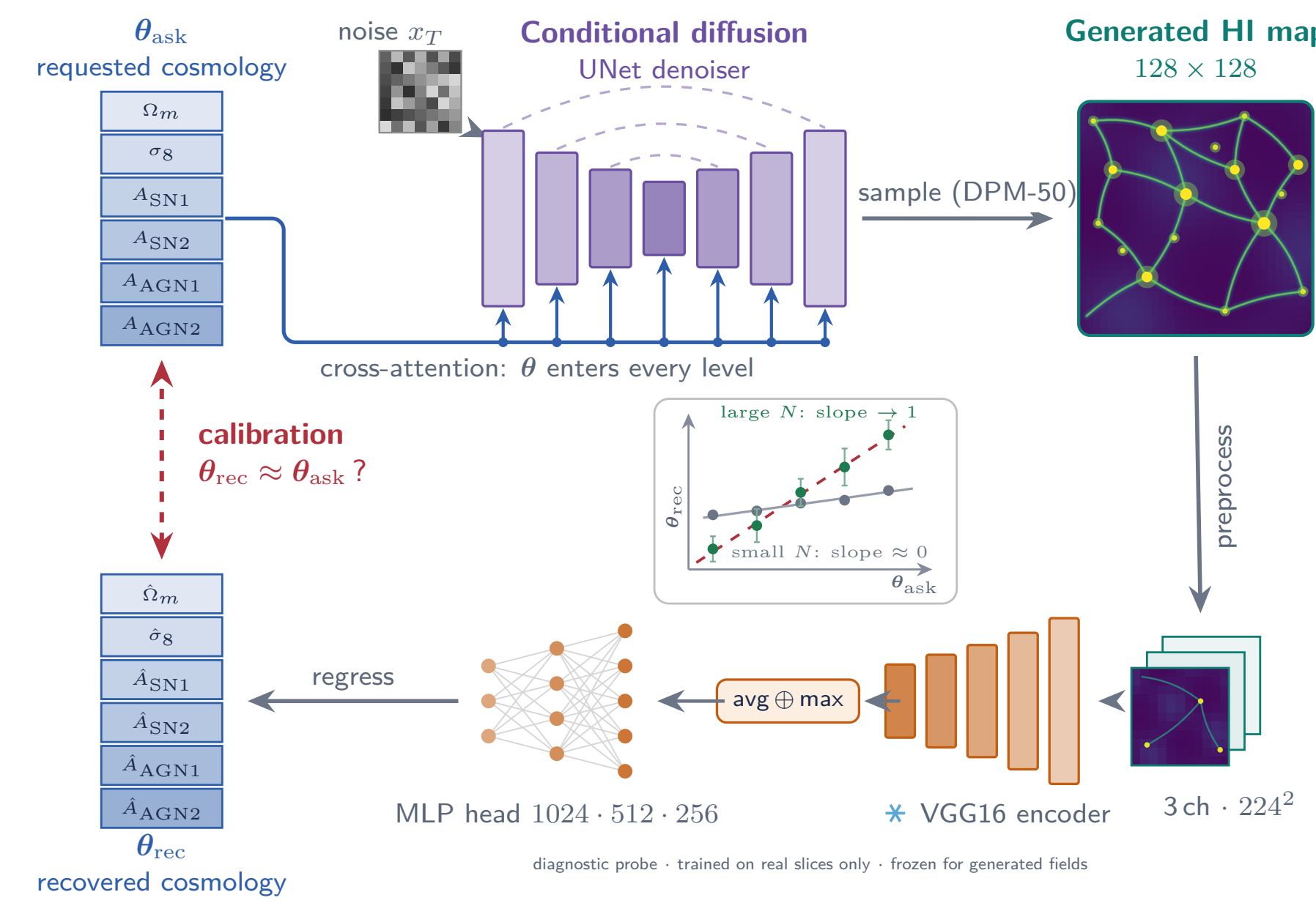
Orange is the true field distribution, blue is the data constraint y , and black points are training-like islands learned under memorization. A likelihood using that collapsed generator can return a sharp but biased posterior, so we test the failure mode directly:

$$\theta_{\text{ask}} \rightarrow x_{\text{gen}} \rightarrow \theta_{\text{rec.}}$$



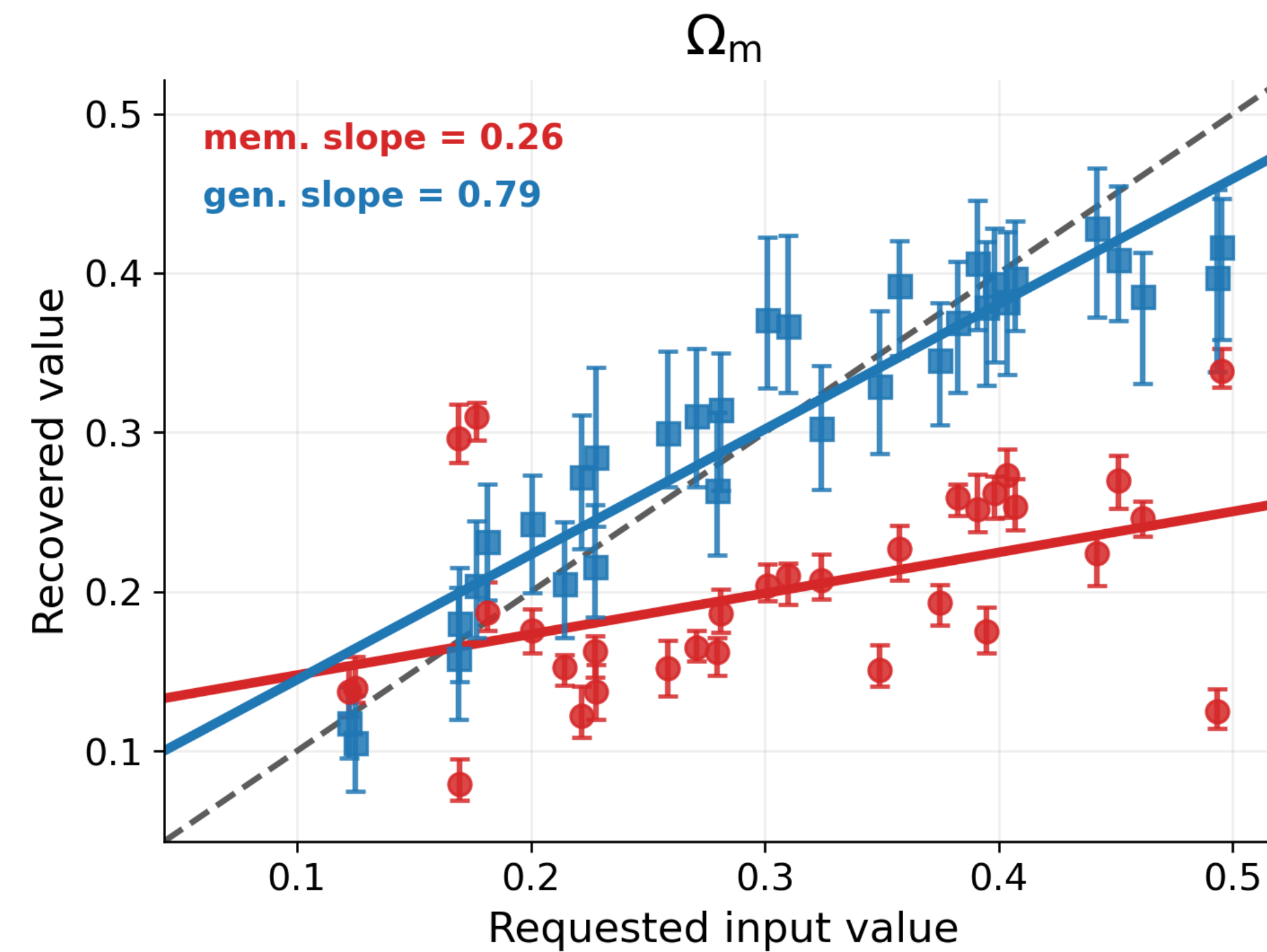
Calibration: does the generator obey θ ?

Ask the conditional model for **held-out cosmologies**, then read the cosmology back from its generated maps with a **frozen probe**. A memorized model returns the training distribution no matter what you ask for; a generalizing model tracks the request.



Probe safeguards: VGG16 frozen (ImageNet); regression head trained on *real* slices from non-held-out simulations only; held-out sims 900–931 used exclusively for testing and calibration.

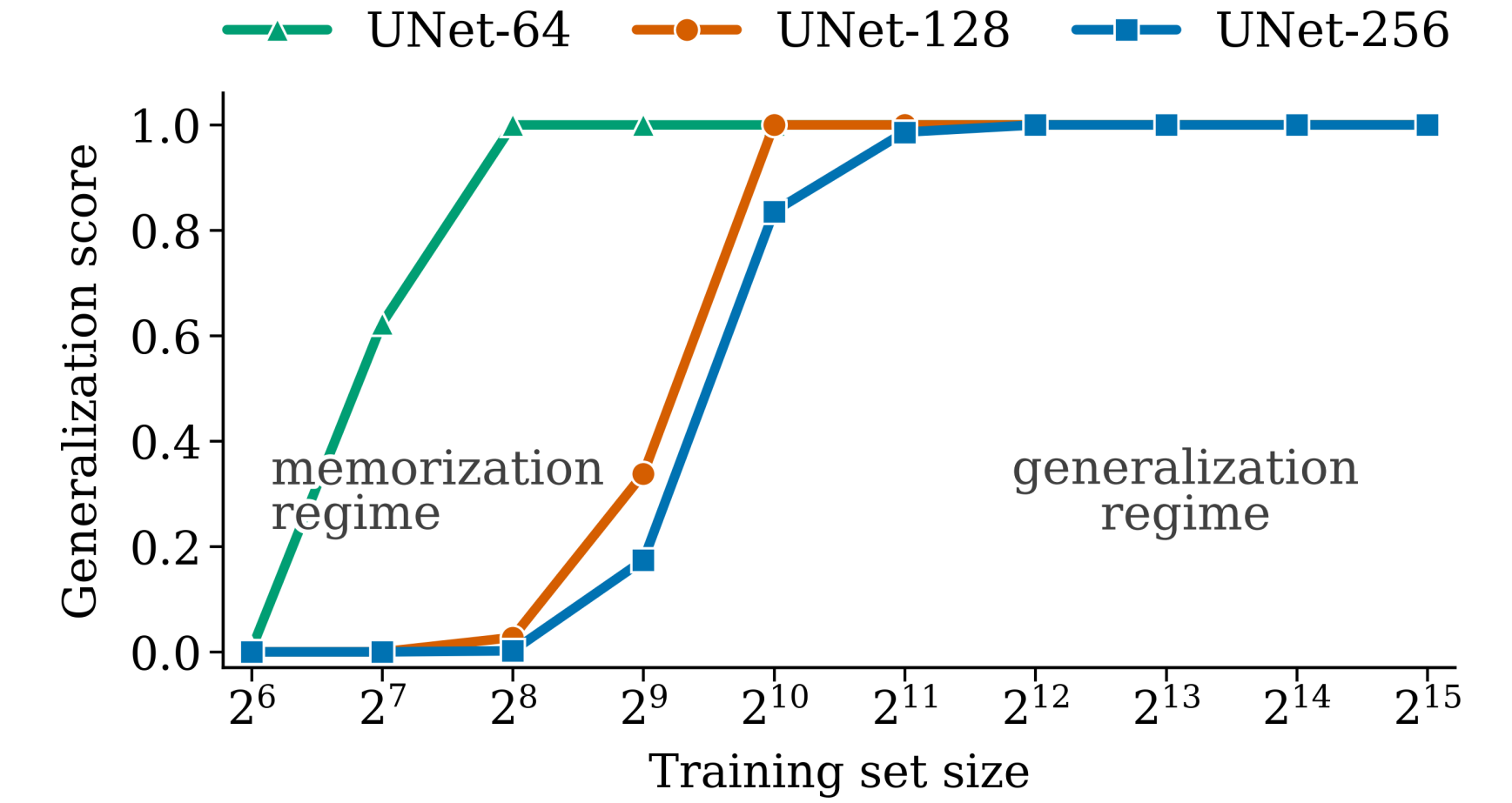
Calibration results



Main result: Ω_m recovery is much flatter under memorization (**slope 0.26**, $N_{2D} = 2^7$) than in the high-data regime (**slope 0.79**, $N_{2D} = 2^{14}$). Error bars are the 16–84 percentile spread over generated seeds at fixed θ_{ask} . We lead with Ω_m because the real-held-out VGG probe is strongest there ($R^2 = 0.91$; $\sigma_8 = 0.74$, feedback parameters weaker).

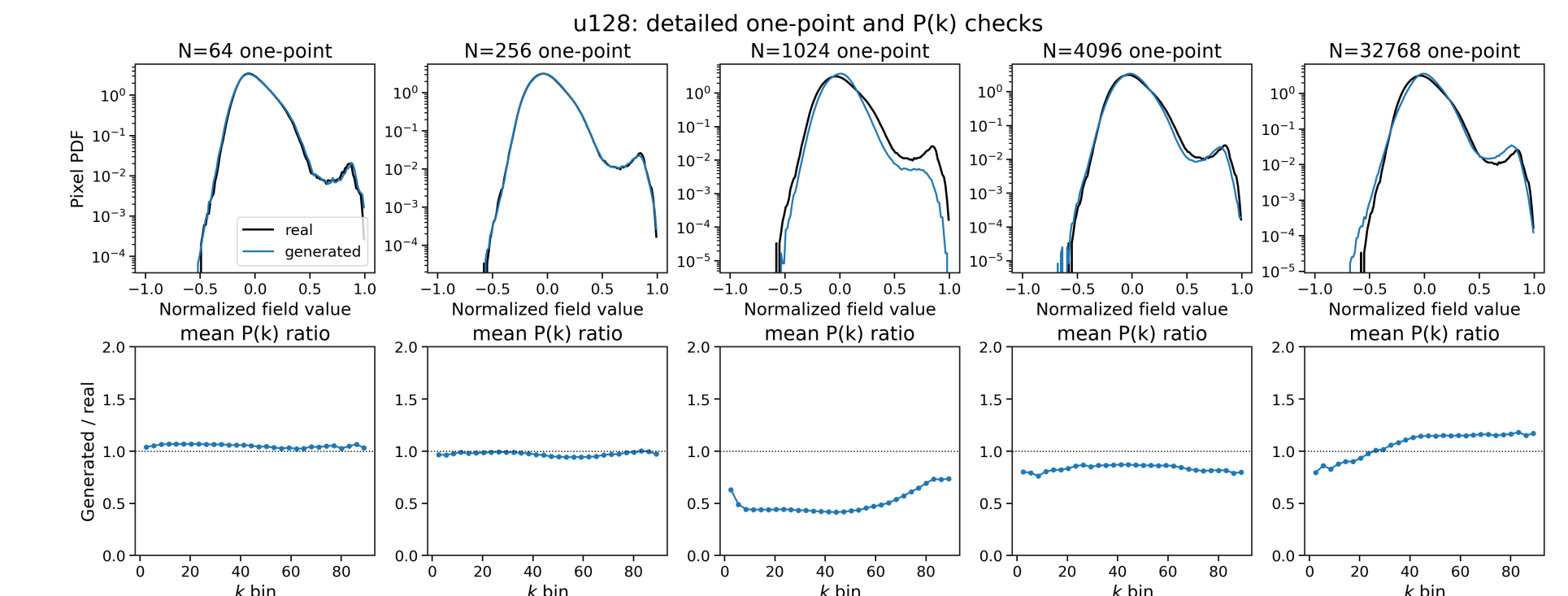
Memorization $\rightarrow$ generalization transition

Generated fields become less training-set-like with more data



- Score uses a **q95-calibrated threshold** ($\tau = 95$ th pct train–train NN similarity): higher means generated fields are less training-set-like in PCA space.
- **Small** N_{2D} : generated maps remain close to training fields.
- **Larger** N_{2D} : UNet-64/128/256 move toward novel samples; the wider UNet-256 transition is shifted to larger data.

Physical fidelity



In high-data examples, the one-point PDFs overlap the real reference and the mean $P(k)$ ratio is mostly within $\sim 20\%$ over the resolved k range. Intermediate undertrained runs visibly fail this check, so novelty alone is not enough: generalization must also preserve field statistics.

Takeaways

1. **Memorization is detectable.** Nearest-neighbor and reproducibility tests locate a data-size transition in CAMELS HI maps.
2. **Conditioning is recoverable.** High-data generated fields track θ_{ask} much better than memorization-regime fields.
3. **Fidelity requires generalization.** Useful emulators must be novel, statistically faithful, and calibrated before inference use.